STUDENT PERFORMANCE & BEHAVIOR ANALYSIS

A Python-Based Data Analytics Project to Explore Factors Associated with Academic Performance

PROJECT REPORT

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Tools: Python | Pandas | NumPy | Matplotlib | Seaborn | Plotly

Based on the project Python notebook and final project presentation

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Executive Summary

This project analyzes 5,000 student records across 23 attributes to identify the demographic, academic and behavioral factors associated with academic performance. The raw data was cleaned (1,025 missing values handled, no duplicates, no invalid values), enriched with six derived columns (29 columns after

cleaning), and explored through univariate, bivariate and multivariate analysis, pivot tables, correlation analysis and statistical summaries. Twenty-one dedicated visualizations were produced.

The main finding is that **assessment components explain performance far better than demographics or lifestyle**. Projects_Score (r = 0.60) and Final_Score (r = 0.59) have the strongest correlations with Total_Score. Gender, department, family income, parent education, attendance, study hours, stress and sleep all show little or no relationship with overall scores. A small but notable subgroup of 313 students (6.26%) combines poor sleep with high stress and may warrant support.

1. Problem Definition & Dataset Selection

1.1 Project Introduction

Student academic performance is influenced by several factors such as study habits, attendance, previous academic performance and behavioral characteristics. This project analyzes student performance and behavioral data to identify important patterns, relationships and factors associated with academic outcomes. The domain is **Education (student performance dashboard)**, one of the sample domains suggested in the project guidelines.

1.2 Problem Statement

To analyze student demographic, academic and behavioral characteristics and identify the key factors associated with student academic performance.

1.3 Objectives

1. Understand the structure and quality of the student dataset.
2. Clean and preprocess the data for reliable analysis.
3. Explore demographic, academic and behavioral patterns.
4. Identify relationships between student characteristics and academic performance.
5. Generate actionable insights and recommendations based on the findings.

1.4 Key Analytical Questions

1. Which academic factors are most closely related to overall student performance?
2. How does academic performance vary across departments and gender?
3. Do attendance and study hours show a meaningful relationship with overall performance?
4. Are there specific student groups that require closer analysis based on stress, sleep or performance?
5. What patterns can be identified to better understand student performance?

1.5 Dataset Information

Item	Detail
Dataset name	Students Performance Dataset
Source	Kaggle (mahmoudelhemaly/students-grading-dataset)

Item	Detail
Link	https://www.kaggle.com/datasets/mahmoudelhemaly/students-grading-dataset/data
Records	5,000 rows (guideline minimum: 500)
Features	23 columns (guideline minimum: 10), a mix of numerical and categorical
Total values	115,000

The 23 original columns fall into three groups:

Group	Columns
Academic	Midterm_Score, Final_Score, Assignments_Avg, Quizzes_Avg, Projects_Score, Participation_Score, Total_Score, Grade
Behavioral	Study_Hours_per_Week, Attendance (%), Stress_Level (1-10), Sleep_Hours_per_Night, Extracurricular_Activities
Background and identity	Student_ID, First_Name, Last_Name, Email, Gender, Age, Department, Parent_Education_Level, Family_Income_Level, Internet_Access_at_Home

2. Methodology

The analysis followed the end-to-end workflow set out in the guidelines:

1. **Load and overview:** import with Pandas; inspect shape, data types, head/tail, info() and describe().
2. **Pre-process:** missing values, duplicates, data types, validation, outliers, derived columns, filtering and aggregation.
3. **Explore:** univariate, bivariate and multivariate analysis using groupby, pivot tables, correlation and statistical summaries.
4. **Visualize:** Matplotlib, Seaborn and Plotly charts with titles, labels, legends and colour schemes.
5. **Interpret:** Markdown explanation of each step, key insights, recommendations and next steps.

3. Data Loading & Initial Overview

The CSV was loaded with `pd.read_csv()`. The dataset contains **5,000 rows and 23 columns (115,000 values)**, using about 898.6 KB of memory. Each row is one student.

Data types: the columns are a combination of text (object), integer and float types. Age and Stress_Level are integers; attendance, scores, study hours and sleep hours are floats; identifiers, gender, department, grade, extracurricular, internet access, parent education and family income are text.

Initial observations from describe():

Measure	Mean
Age	21.05 years
Attendance	75.36%
Midterm score	70.70
Final score	69.55
Total score	71.65
Study hours per week	17.52
Stress level (1-10)	5.51
Sleep per night	6.51 hours

Student_ID and Email are unique for all 5,000 students. Gender has 2 categories, Department has 4 (Mathematics, Business, Engineering, CS), Grade has 5 (A to F), and Family_Income_Level has 3 (Low, Medium, High). Parent_Education_Level had only 3,975 non-null values, which flagged the one missing-data problem in the dataset.

Categorical distributions:

- Gender: 2,551 male and 2,449 female students.
- Departments are similar in size: Engineering is largest (1,274) and Mathematics smallest (1,223).
- Ages range from 18 to 24, with age 21 the most frequent (753 students).
- Grade C is the most common (2,307), followed by D (1,760); Grade A has only 16 students.
- Extracurricular participation and home internet access are split almost evenly between Yes and No.
- Low family income is the most common income category (1,687 students).

4. Data Pre-processing

4.1 Summary of Cleaning Steps

Check	Result	Action
Missing values	1,025 nulls (20.5%) in Parent_Education_Level only	Filled with a separate category, Unknown
Duplicate rows	0	None needed
Data types	All appropriate	None needed

Check	Result	Action
Range and category validation	No invalid values found	None needed
Outliers	4 IQR outliers in Total_Score	Retained as genuine values

4.2 Missing Values

Only Parent_Education_Level had missing data (1,025 of 5,000 records, 20.5%), visualized with a bar chart of null counts per column. Replacing the gaps with the mode (Bachelor's) would have assumed these students all belong to the most common group and introduced bias. Instead the nulls were filled with **Unknown**, which preserves all 5,000 records and makes the data gap explicit. After treatment, the dataset had zero missing values.

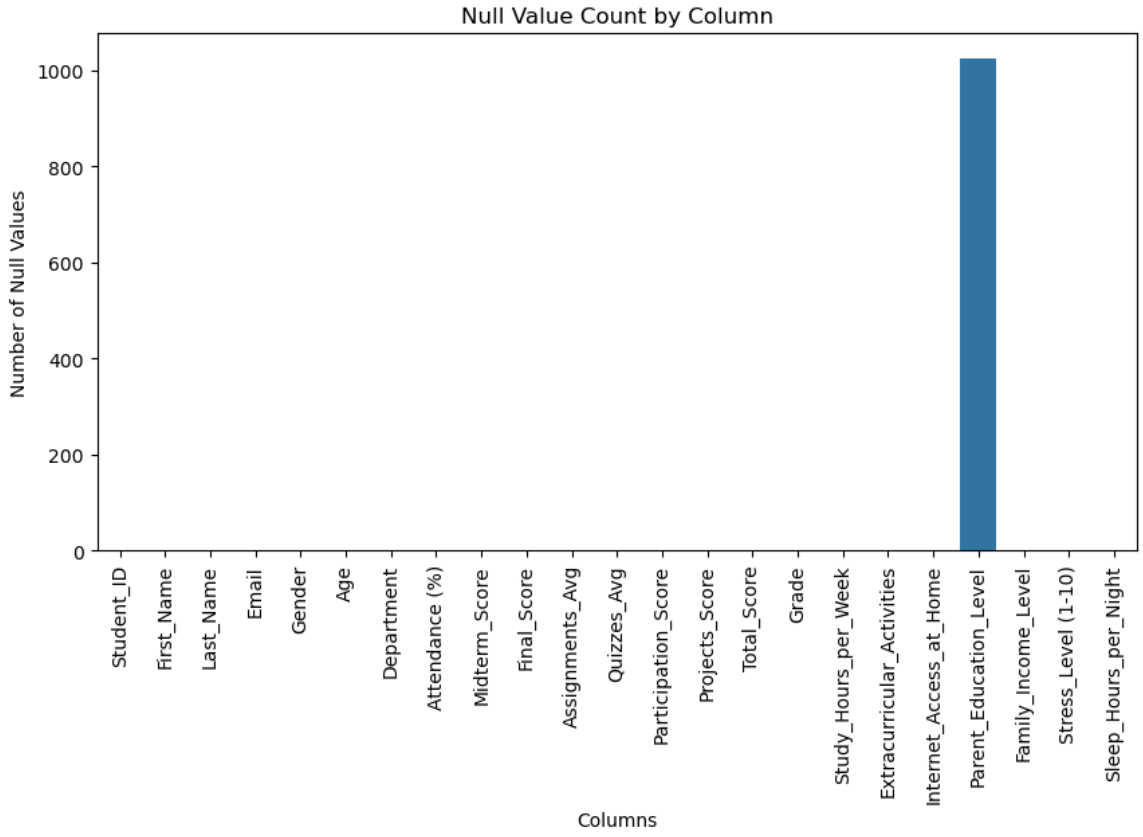


Figure 4.1: Null values per column. Only Parent_Education_Level has missing data (1,025 records).

4.3 Duplicates and Data Types

`df.duplicated().sum()` returned 0, and Student_ID has no repeats, so every row is a distinct student. Data types were reviewed column by column and judged correct, so no conversions were required.

4.4 Data Validation

Rule-based checks were applied to every relevant column and found **no invalid records**:

- Attendance between 0 and 100%; age between 18 and 24; Student_ID unique.
- All seven score columns between 0 and 100.
- Study hours, sleep hours not negative; stress level between 1 and 10.
- Gender, Department, Extracurricular_Activities, Internet_Access_at_Home, Parent_Education_Level and Family_Income_Level contain only expected categories.

4.5 Outlier Detection

A box plot of all numeric columns showed reasonable ranges. Using the IQR rule on Total_Score: Q1 = 66.53, Q3 = 76.71, IQR = 10.18, giving bounds of 51.27 and 91.98. Only **4 observations** fall outside these bounds. They are valid scores within 0 to 100 and represent real variation in performance, so they were retained.

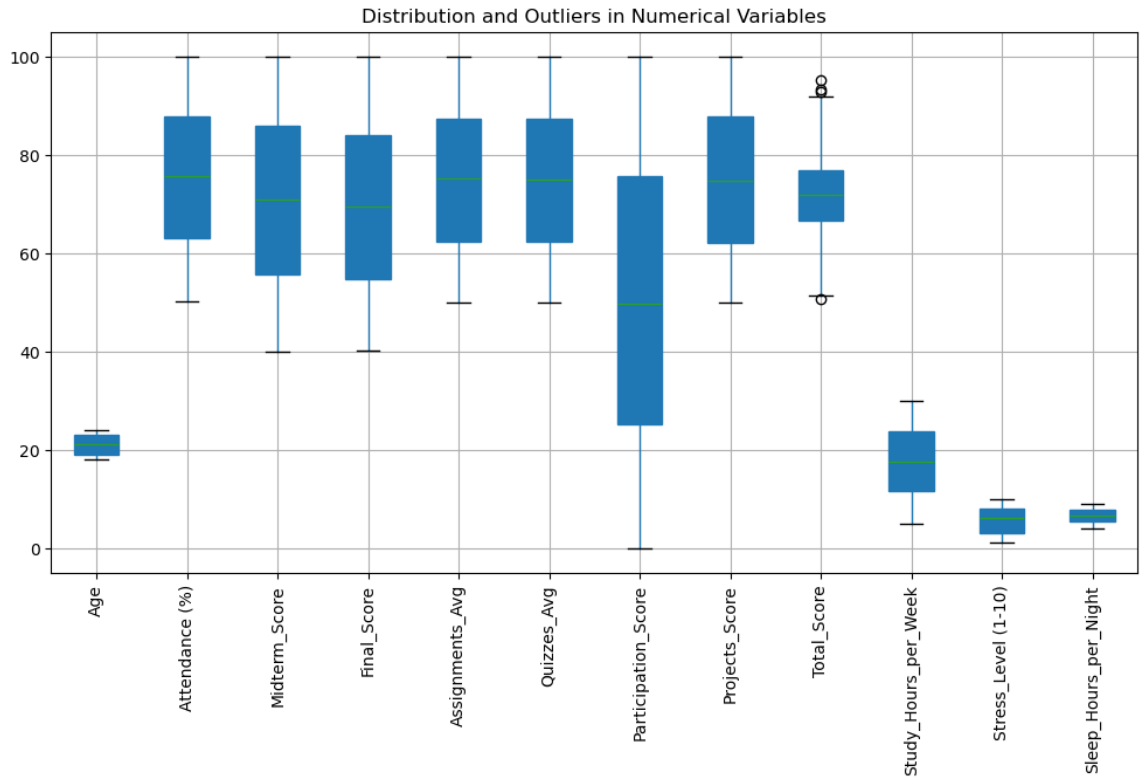


Figure 4.2: Box plots of all numeric columns used for the outlier check.

4.6 Derived Columns

Six columns were created, taking the dataset from 23 to **29 columns**.

New column	Built from	Method and categories
Full_Name	First_Name + Last_Name	String concatenation; single identifier per student
Age_Group	Age	pd.cut: Late Teens (18-19), Early 20s (20-21), Mid 20s (22-24)

New column	Built from	Method and categories
Attendance_Status	Attendance (%)	pd.cut: Low (50-60), Moderate (61-75), Good (76-85), Excellent (86-100)
Performance Category	Grade	Mapping: A Excellent, B Good, C Average, D Poor, F Very Poor
Stress_Category	Stress_Level	pd.cut: Low (1-3), Moderate (4-7), High (8-10)
Sleep_Category	Sleep_Hours_per_Night	pd.cut: Poor (up to 5 h), Average (5 to 7 h), Good (over 7 h)

These categories make groups easier to compare and visualize than raw numbers.

4.7 Filtering

Filter	Condition	Students
Top performers	Total_Score > 90	16
Female students	Gender = Female	2,449
Male students	Gender = Male	2,551
With home internet	Internet_Access_at_Home = Yes	2,520
Without home internet	Internet_Access_at_Home = No	2,480
Low income, highly educated parents	Income = Low and Parent education = Master's or PhD	685
Poor sleep and high stress	Sleep <= 5 h and Stress >= 8	313

4.8 Aggregation

Groupby aggregations were built by department, by grade, by performance category (average sleep) and by gender (average score).

By department:

Department	Students	Avg Total Score
Business	not listed separately	71.92
Engineering	1,274	71.69
Mathematics	1,223	71.55

Department	Students	Avg Total Score
CS	not listed separately	71.43

Business also had the highest average attendance (75.76%).

By grade:

Grade	Avg Total Score	Avg Final Score
A	91.28	95.17
B	83.13	85.23
C	74.52	73.80
D	65.80	61.01
F	57.55	50.88

By gender: female 71.64, male 71.66.

The final check confirmed 5,000 rows, 29 columns, 0 missing values and 0 duplicate rows.

5. Exploratory Data Analysis

5.1 Univariate Analysis

Categorical variables were profiled with counts and percentages (nine variables).

- **Gender:** 51.02% male, 48.98% female, nearly balanced.
- **Department:** Engineering 25.48% down to Mathematics 24.46%; no department dominates.
- **Age group:** Mid 20s is the largest at 43.78%, partly because it spans three ages.
- **Attendance status:** Excellent is the largest category (31.02%); Low is 19.46%.
- **Performance category:** concentrated in Average and Poor; Excellent is only 0.32%.
- **Extracurricular activities and family income:** roughly even splits.
- **Stress:** most students fall in the Moderate or High categories.
- **Sleep:** 79.30% have Average or Good sleep; 20.70% have Poor sleep (about one in five).

Numerical variables were summarized with mean, median, standard deviation, minimum, maximum and quartiles. Total Score averages about 72, attendance about 75%, study time about 17.5 hours a week, sleep about 6.5 hours a night and stress about 5.5 out of 10. Assignments, quizzes and projects all average around 75. Participation shows the highest variation among academic measures.

5.2 Bivariate Analysis

Categorical vs numerical

Relationship	Result
Grade vs Total Score	Falls steadily from 91.28 (A) to 57.55 (F)

Relationship	Result
Department vs Total Score	71.43 to 71.92; very close
Gender vs Total Score	71.64 (F) vs 71.66 (M); virtually identical
Parent education vs Total Score	71.45 (High School) to 71.98 (Bachelor's); Unknown 71.58
Family income vs Total Score	High 71.71, Low 71.71, Medium 71.54
Grade vs Attendance	Does not rise with grade: A 72.86%, F 76.80%
Performance vs Stress	5.42 (Very Poor) to 5.75 (Excellent); small spread
Performance vs Sleep	6.48 h (Very Poor) to 6.92 h (Excellent); small spread

Numerical vs numerical

- Study Hours vs Total Score: r = -0.01 (almost no linear relationship).
- Attendance vs Total Score: r = -0.01 (almost no linear relationship).
- Projects and Final scores show a strong positive relationship with Total Score; Midterm and Assignments are moderate; Quizzes and Participation are weak.

5.3 Multivariate Analysis

- **Department, gender and score (pivot):** male and female averages differ by well under half a point within every department (for example CS 71.43 vs 71.44, Business 71.95 vs 71.89).
- **Department, grade and count (pivot):** Grade C is the most common in every department and Grade A the rarest (Business 2, Engineering 4, CS 5, Mathematics 5).
- **Stress, sleep and performance (groupby):** the Excellent group has the highest average sleep (6.92 h) and stress (5.75), while stress is otherwise nearly flat across groups. The Excellent group has only 16 students, so it should be read with caution.

Overall, department, gender, stress and sleep have little effect on scores in this dataset.

5.4 Groupby and Pivot Table Summaries

Five groupby summaries and five pivot tables were built. The patterns are consistently flat:

- Average Final Score by department ranges from 69.01 (Mathematics) to 70.09 (Business), a gap of 1.08 points.
- Average study hours by department range from 17.22 (CS) to 17.76 (Engineering), a gap of 0.54 hours.
- Average sleep is 6.57 h for males and 6.45 h for females, about 7 minutes apart.
- By grade, study hours range only from 17.31 to 18.73 and attendance from 72.86% to 76.80%, while Final Score swings from 95.17 to 50.88.
- Department averages for stress (5.44 to 5.59) and sleep (6.48 to 6.53) are nearly identical.
- Pivot tables on attendance by department and internet access, total score by gender and extracurricular activity, and stress and sleep by age group and gender all showed small

differences with no consistent pattern. Males report slightly higher stress and sleep in older age groups.

5.5 Correlation Analysis

A correlation matrix of 12 numerical variables (Age, Attendance, Midterm, Final, Assignments, Quizzes, Participation, Projects, Total Score, Study Hours, Stress, Sleep) was computed and shown as a heatmap.

Strength	Variables related to Total Score
Strongest	Projects_Score (0.60), Final_Score (0.59)
Moderate	Midterm_Score, Assignments_Avg
Weak	Quizzes_Avg, Participation_Score
Near zero	Age, Attendance, Study Hours, Stress, Sleep

Scores depend more on academic assessments than on behavioral factors.

5.6 Statistical Summaries

- **Mean vs median:** nearly equal for most variables, suggesting balanced, roughly symmetric distributions.
- **Standard deviation:** Participation_Score has the highest (28.99); Total_Score is relatively low (7.23).
- **Coefficient of variation:** participation, stress and study hours vary most in relative terms; Total Score and Age vary least. Behavioral variables vary more than academic outcomes.
- **Quartiles and IQR:** the middle 50% of Total Scores lie between 66.53 and 76.71 (IQR 10.18); 4 outliers, all retained.
- **Percentage findings:** 16 students (0.32%) scored above 90; 313 (6.26%) have poor sleep with high stress; 685 (13.70%) are from low-income families with a Master's or PhD parent.

6. Visualizations

Twenty-one visualizations were produced in the dedicated section, plus two during pre-processing (null-value bar chart and numeric box plot), exceeding the guideline minimum of 10. All use Matplotlib, Seaborn or Plotly, with titles, axis labels, legends and colour schemes. They include subplots, a pairplot and two interactive Plotly charts.

No.	Visualization	Chart type	Key observation
6.1	Students by grade	Bar plot	C is the largest group, then D and B; A and F are small
6.2	Gender distribution	Pie chart	Nearly balanced, slightly more male
6.3	Students by	Horizontal bar	Engineering highest,

No.	Visualization	Chart type	Key observation
	department		Mathematics lowest, small gap
6.4	Age distribution	Histogram	Fairly even across ages 18 to 24
6.5	Total Score density	KDE plot	Peak around 70 to 75; range from low 50s to mid 90s
6.6	Total Score by grade	Violin plot	Clear separation: about 91, 83, 75, 66 and 58 from A to F
6.7	Average score by department	Bar plot	Business 71.92 vs CS 71.43
6.8	Attendance vs Total Score	Scatter plot	No clear trend; horizontal grade bands
6.9	Study hours vs Total Score	Regression plot	Almost flat regression line
6.10	Average score across grades	Line plot	Consistent decline from A to F
6.11	Stress level distribution	Count plot	Students spread across all ten levels
6.12	Sleep vs stress	Scatter plot	No strong linear relationship
6.13	Average score by gender	Bar plot	Difference of only 0.02 points
6.14	Correlation heatmap	Heatmap	Assessment variables dominate the correlations with Total Score
6.15a	Stress and sleep by grade	Grouped bar	Stress 5.42 to 5.75; sleep 6.48 to 6.92 h
6.15b	Stress category by grade	Stacked percentage bar	Stress mix is similar in every grade
6.16	Academic measures	Box plots	Comparable spreads for the score

No.	Visualization	Chart type	Key observation
			components
6.17a	Midterm, final, attendance, study hours	2x2 subplot of histograms	Wide variation rather than a single concentration
6.17b	Pairwise relationships	Pairplot	Clearest links among Midterm, Final and Total Score
6.18a	Study hours vs Total Score	Interactive Plotly scatter	Confirms no upward trend; hover shows student details
6.18b	Average score by department	Interactive Plotly bar	Range of 71.43 to 71.92

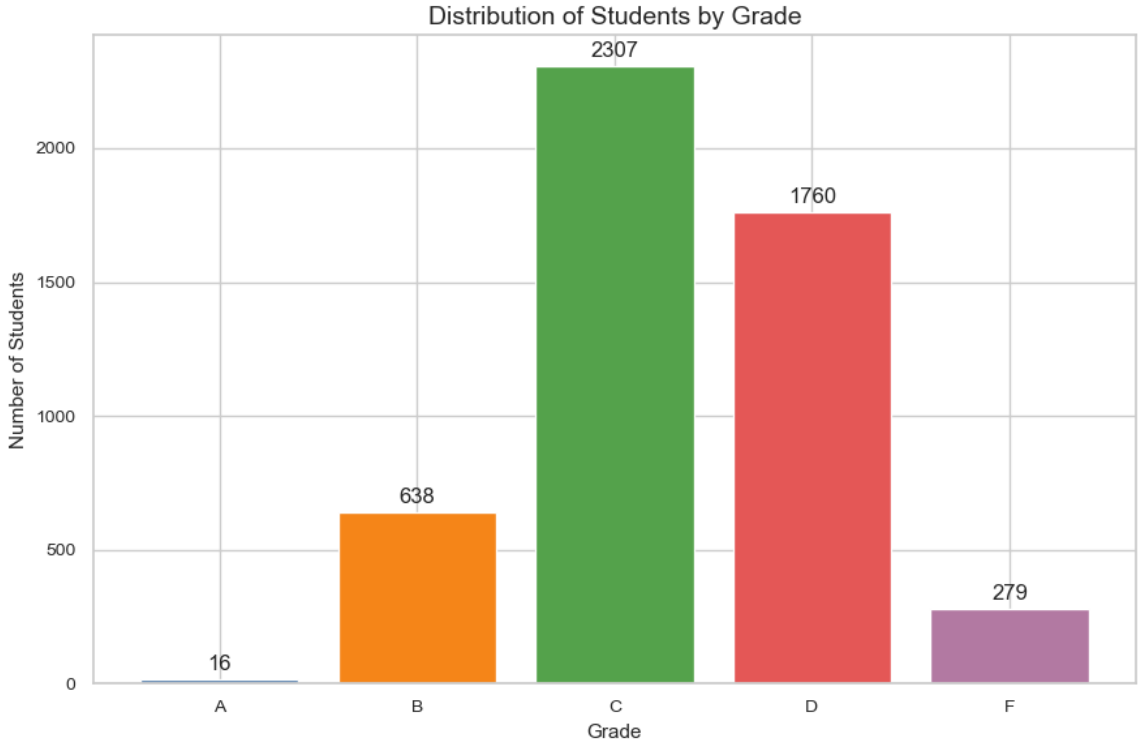


Figure 6.1: Students by grade (bar plot).

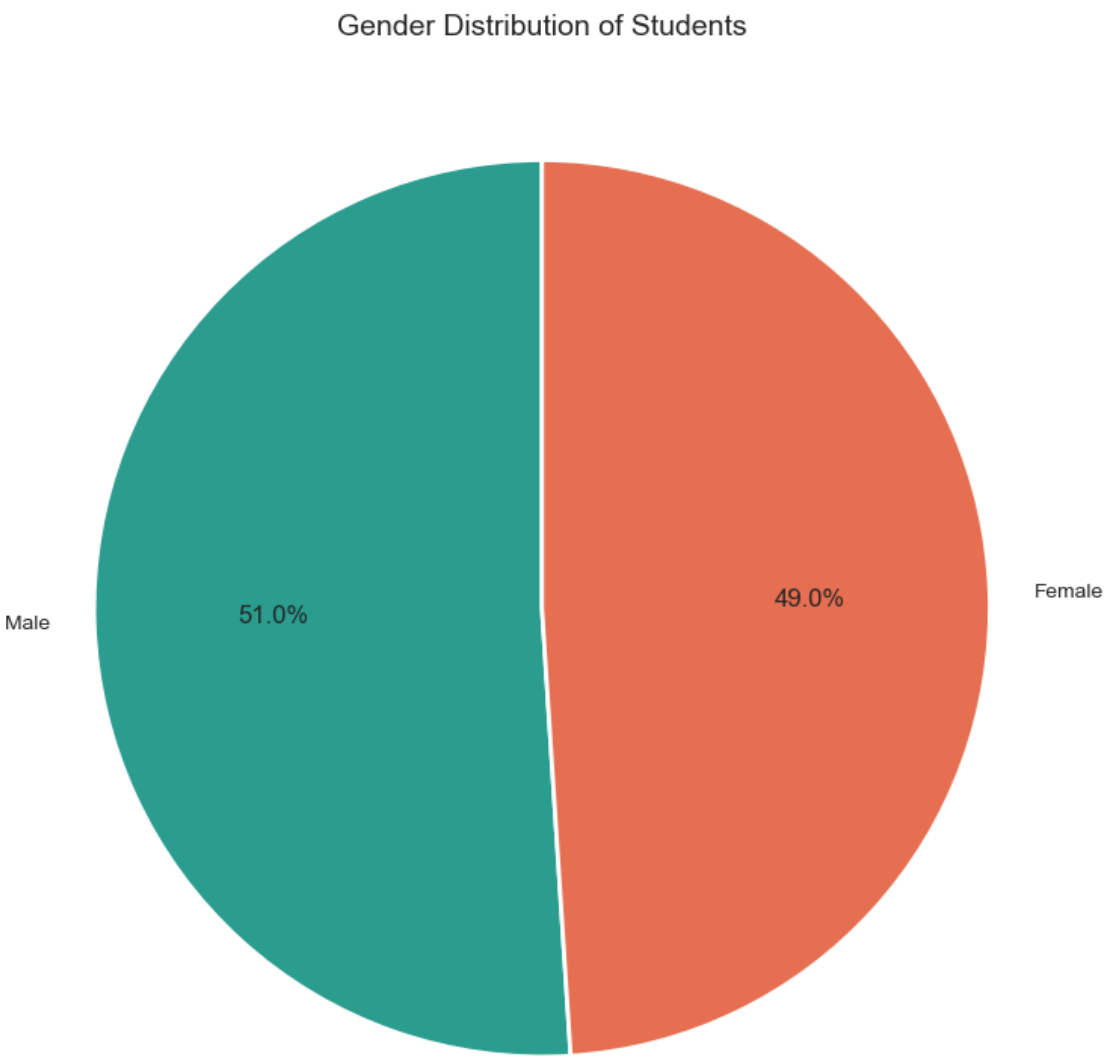


Figure 6.2: Gender distribution (pie chart).

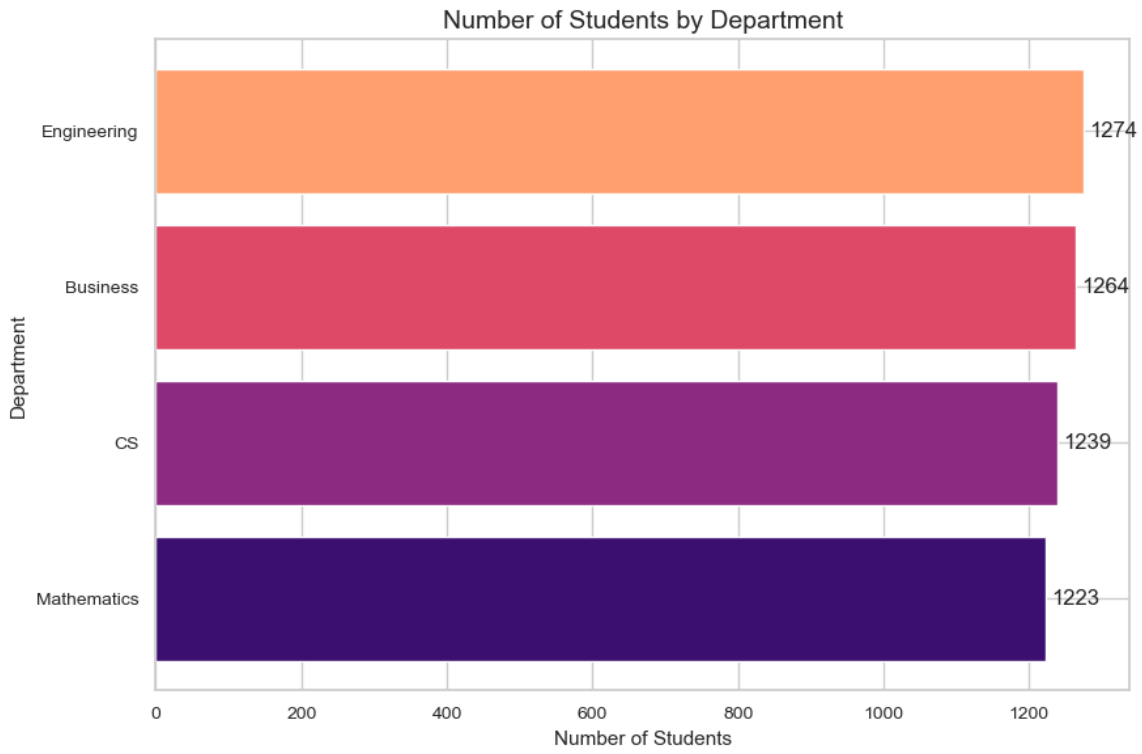


Figure 6.3: Students by department (horizontal bar chart).

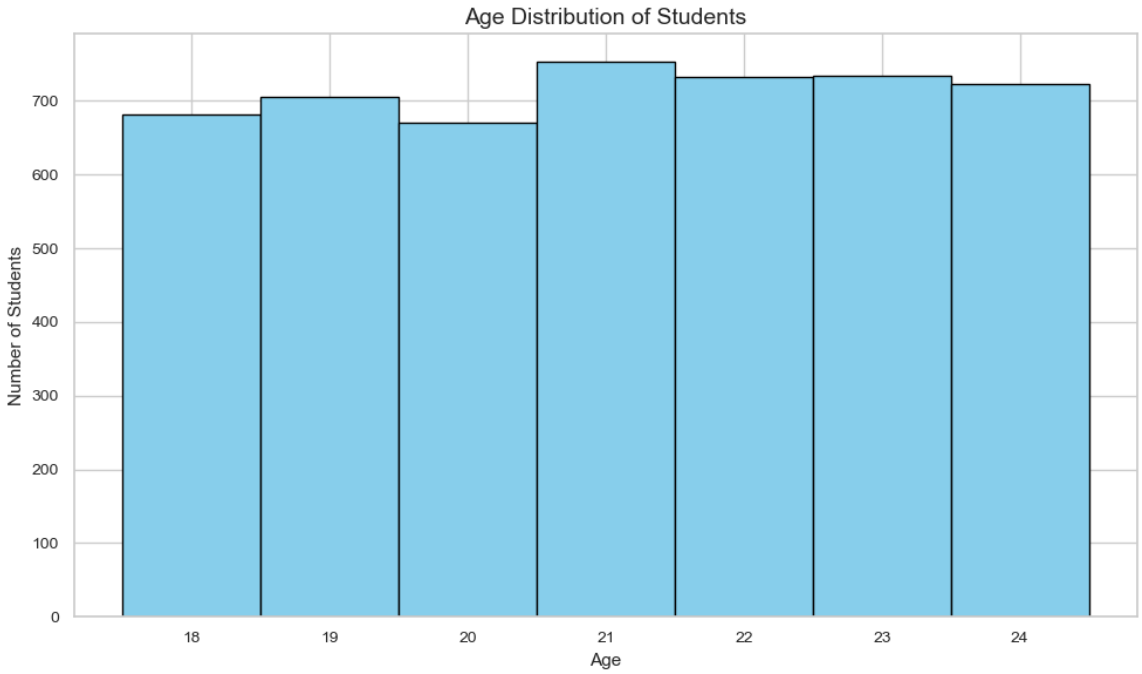


Figure 6.4: Age distribution (histogram).

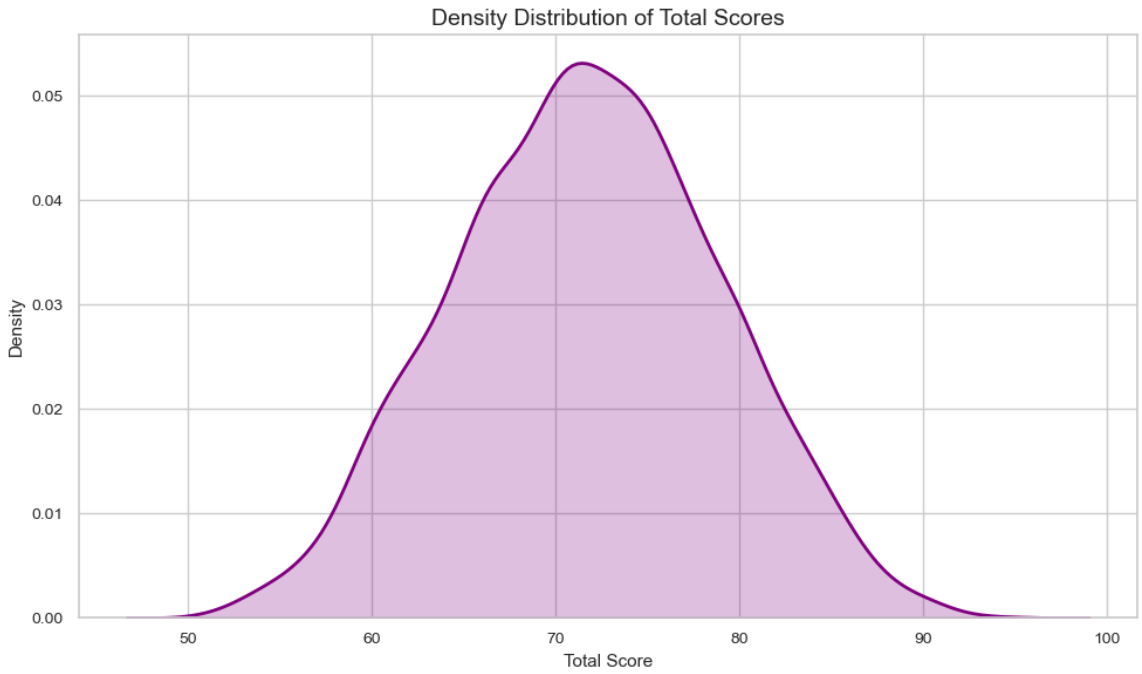


Figure 6.5: Total Score density (KDE plot).

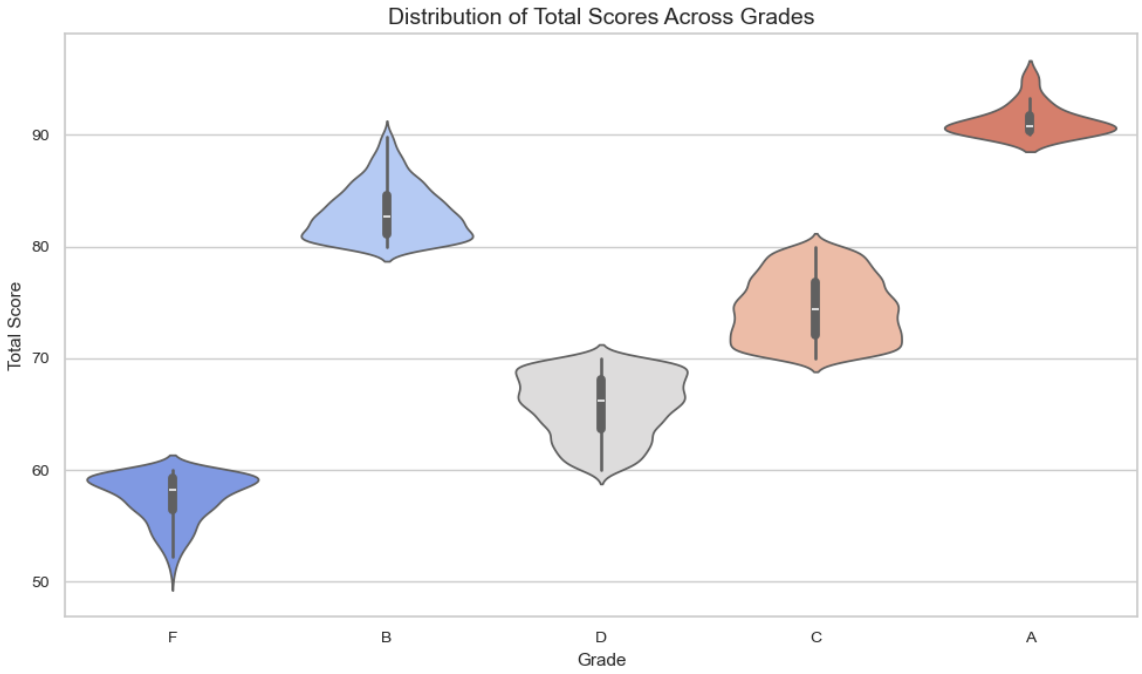


Figure 6.6: Total Score by grade (violin plot).

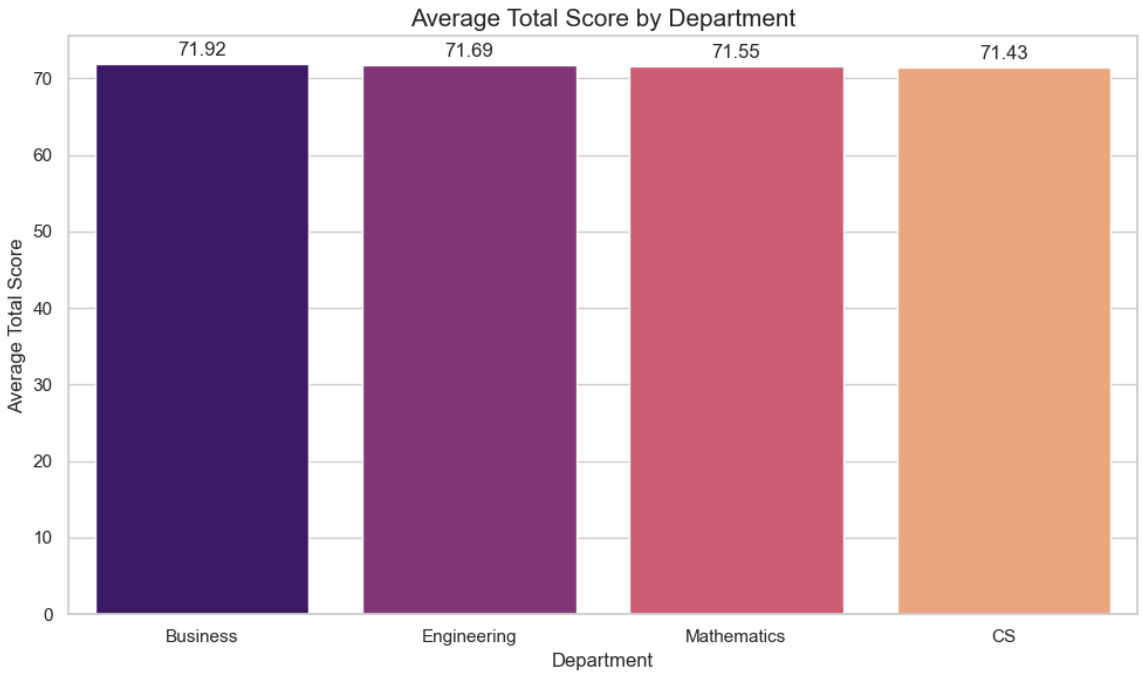


Figure 6.7: Average Total Score by department (bar plot).

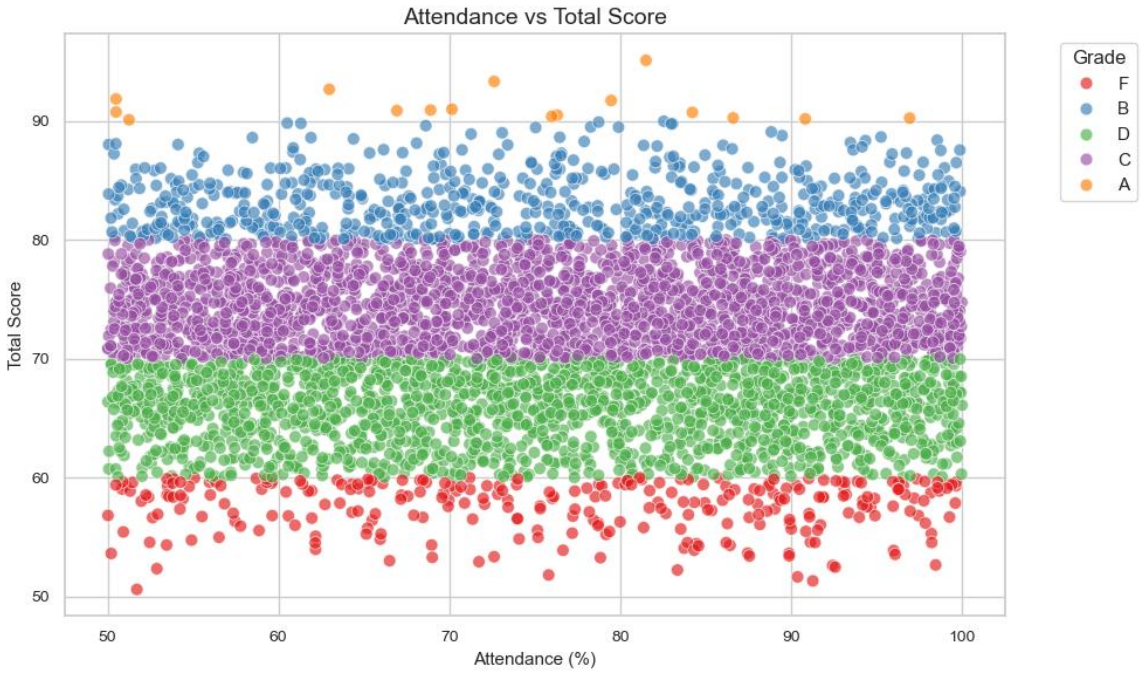


Figure 6.8: Attendance vs Total Score (scatter plot).

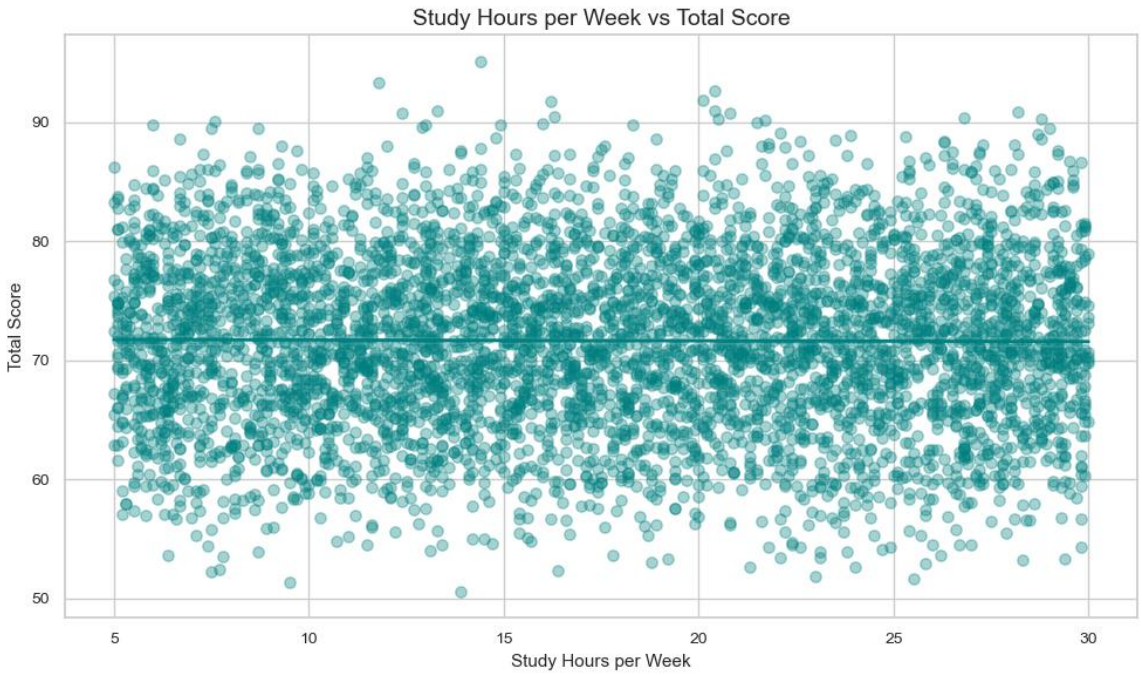


Figure 6.9: Study hours vs Total Score (regression plot).

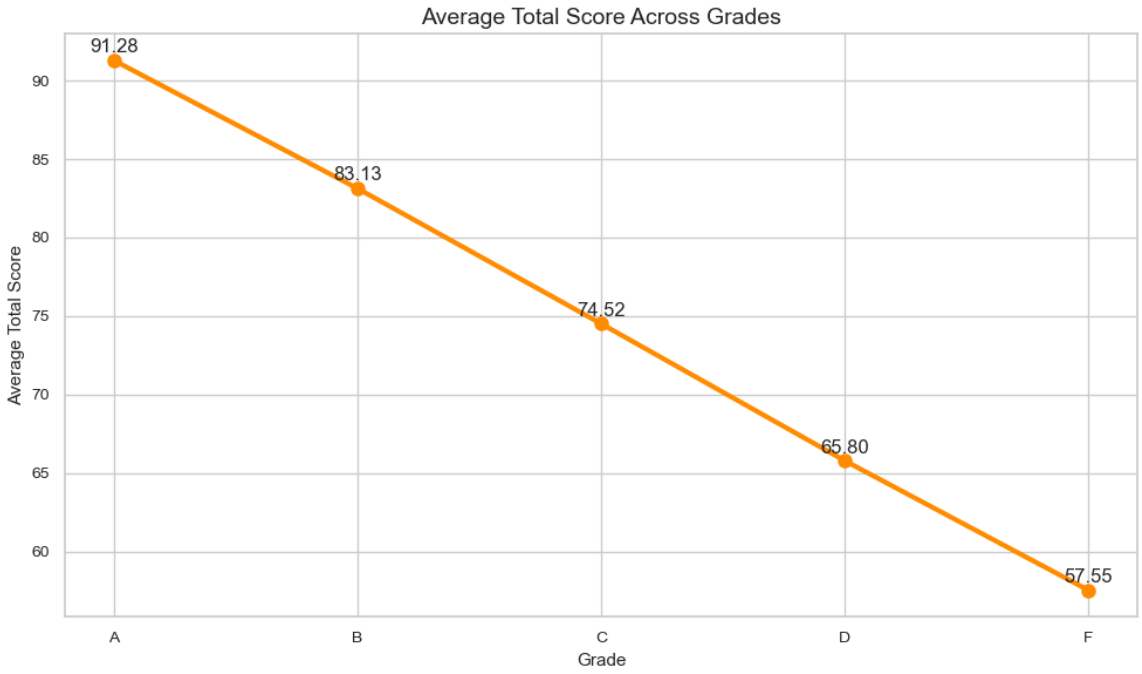


Figure 6.10: Average Total Score across grades (line plot).

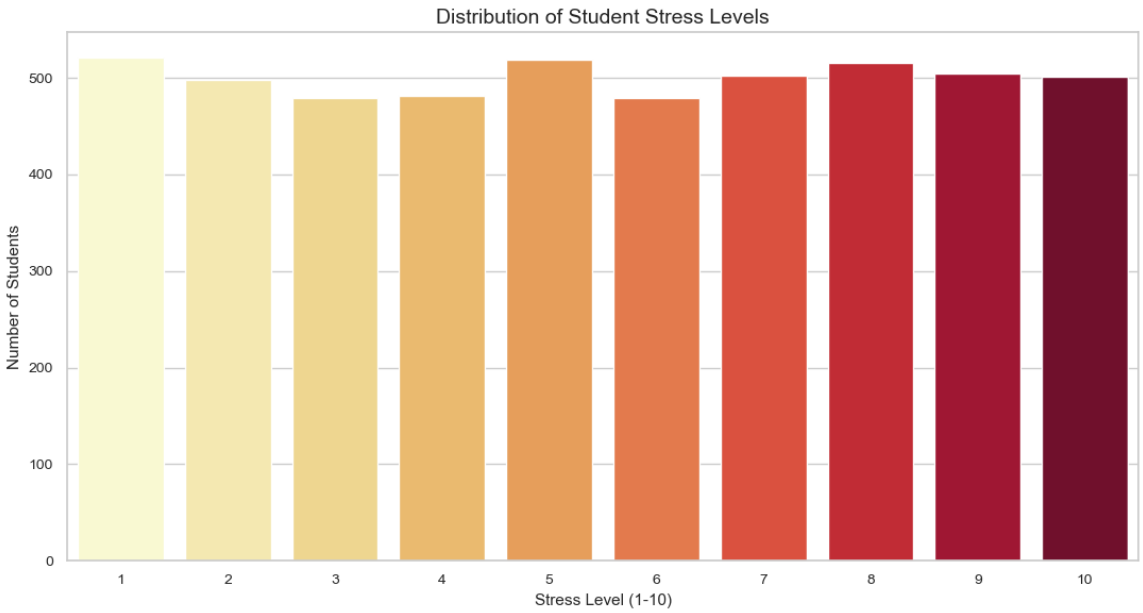


Figure 6.11: Stress level distribution (count plot).



Figure 6.12: Sleep hours vs stress level (scatter plot).

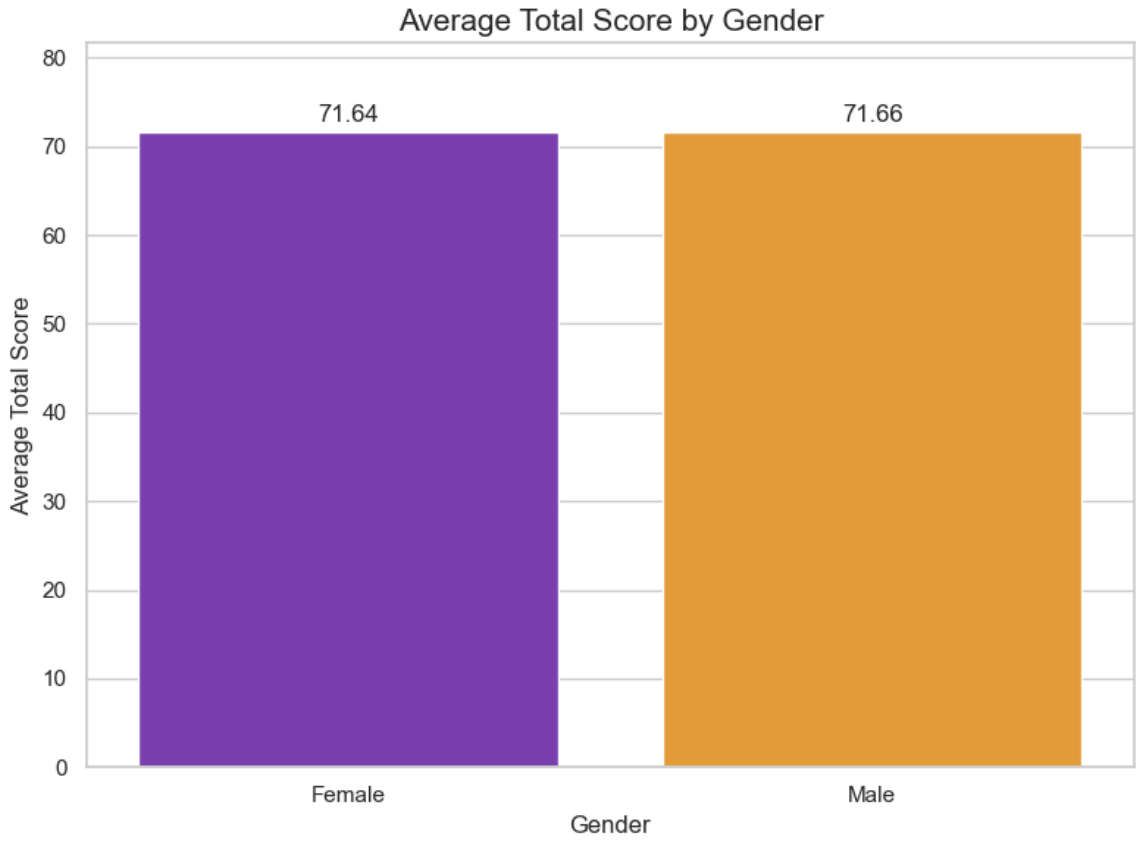


Figure 6.13: Average Total Score by gender (bar plot).

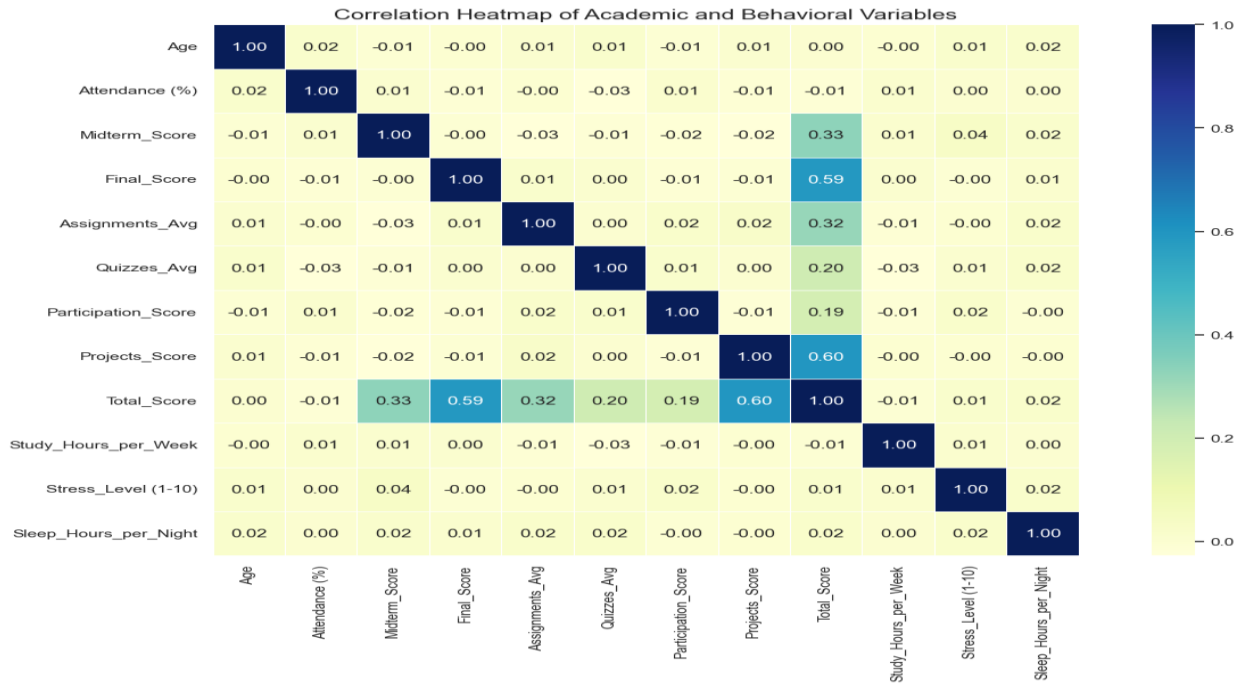


Figure 6.14: Correlation heatmap of the numerical variables.

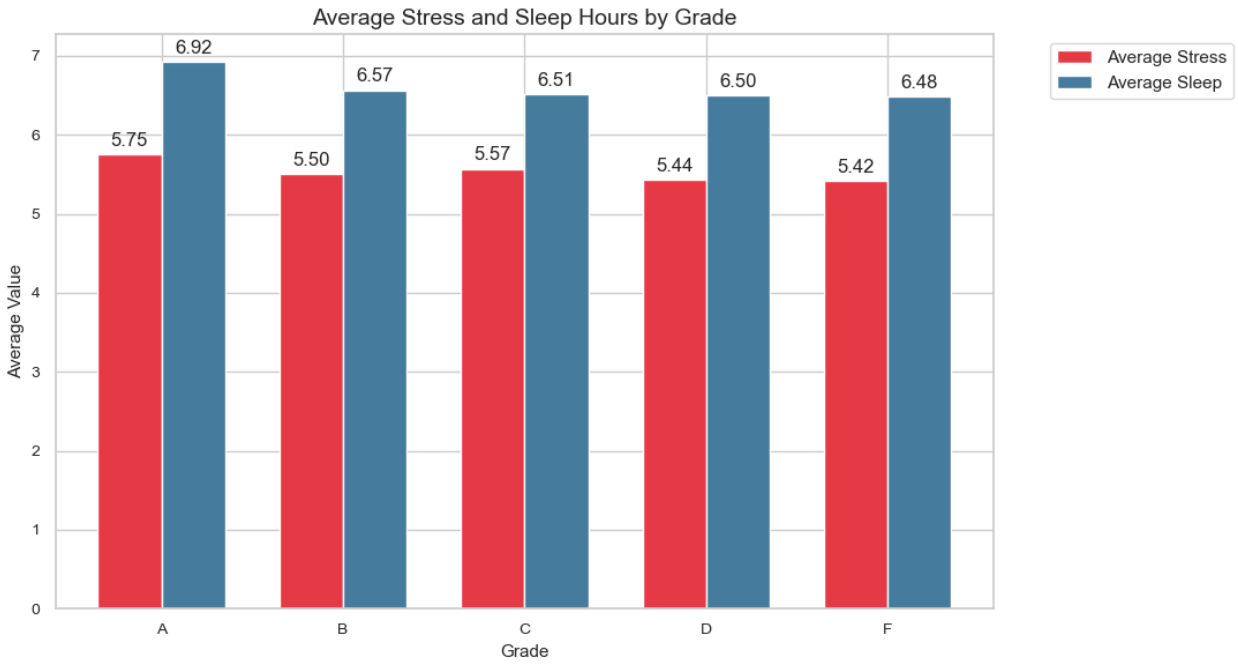


Figure 6.15a: Average stress and sleep by grade (grouped bar chart).



Figure 6.15b: Stress category share within each grade (stacked percentage bar chart).

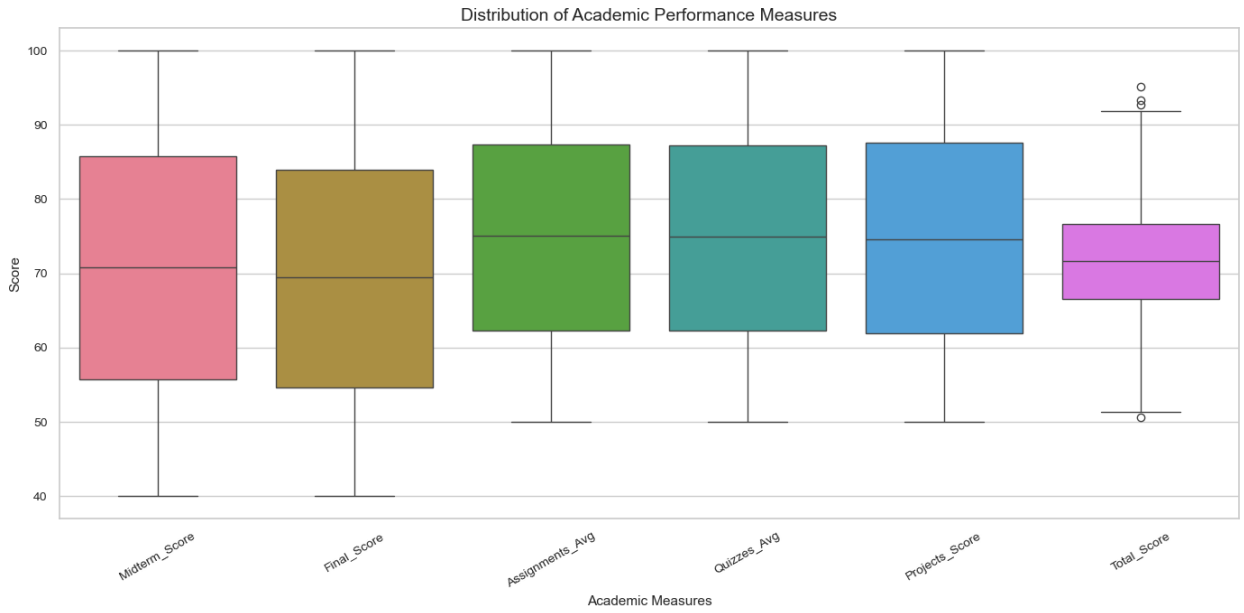


Figure 6.16: Academic score components (box plots).

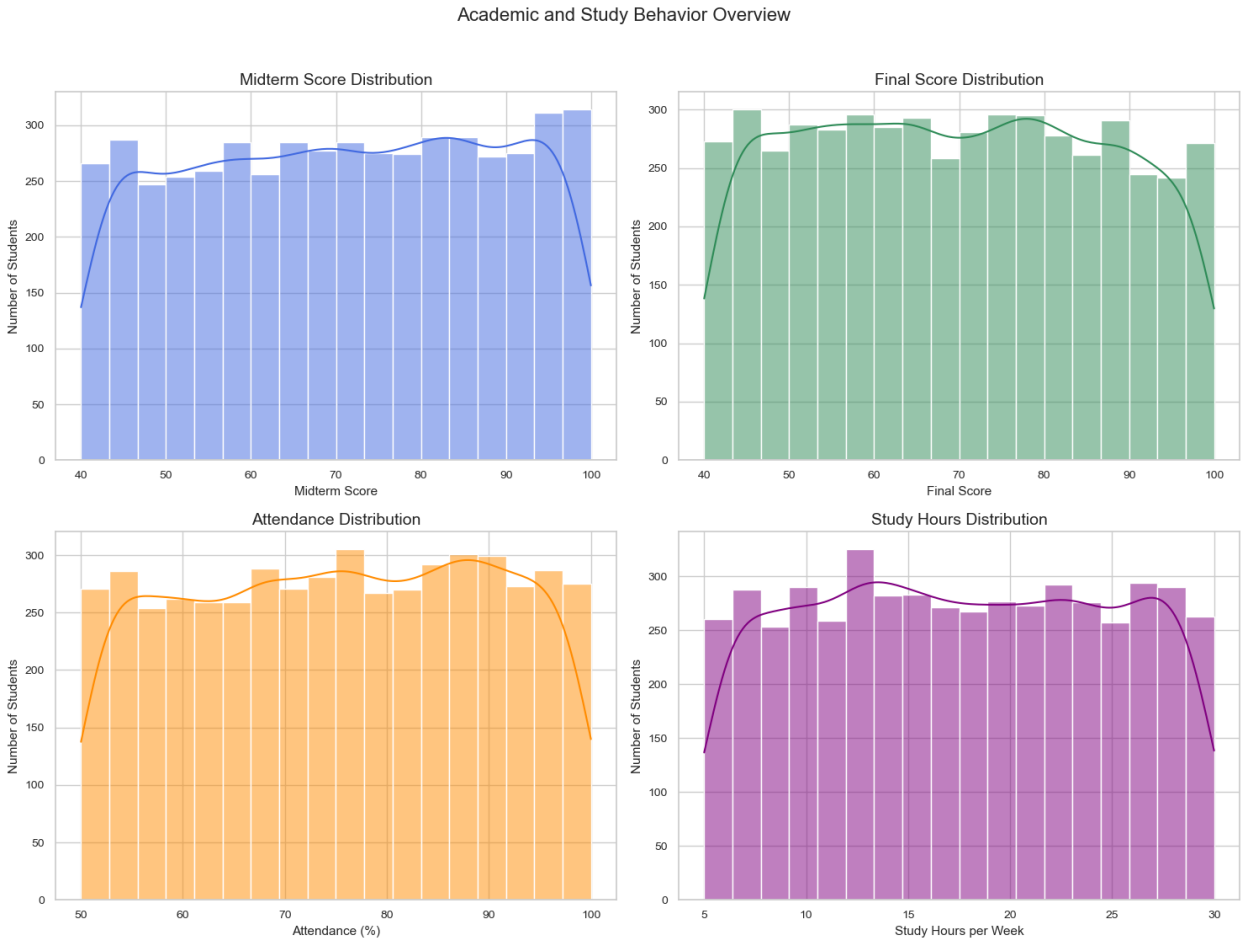


Figure 6.17a: Midterm, Final, Attendance and Study Hours distributions (2x2 subplot).

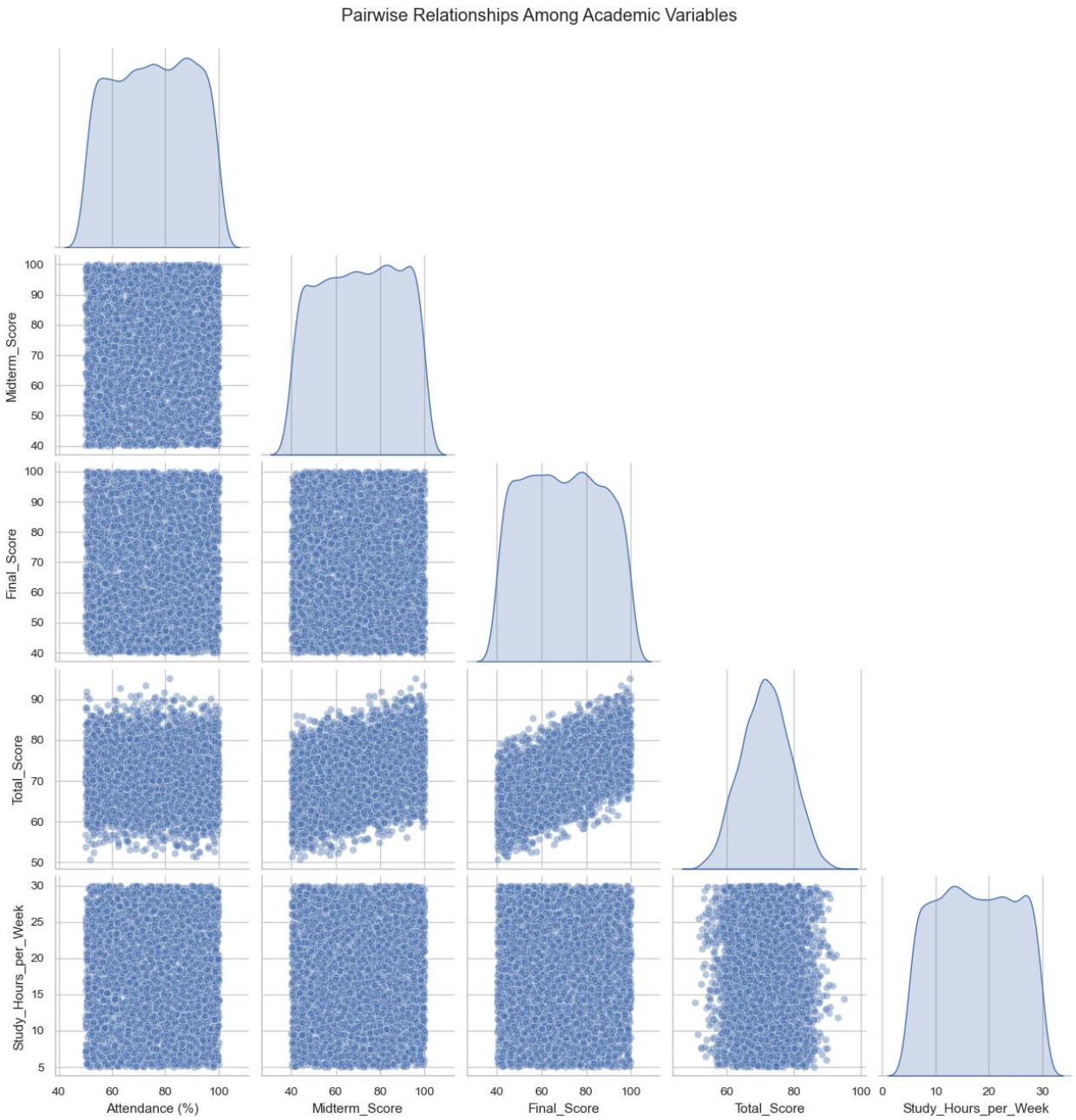


Figure 6.17b: Pairwise relationships between key variables (pairplot).

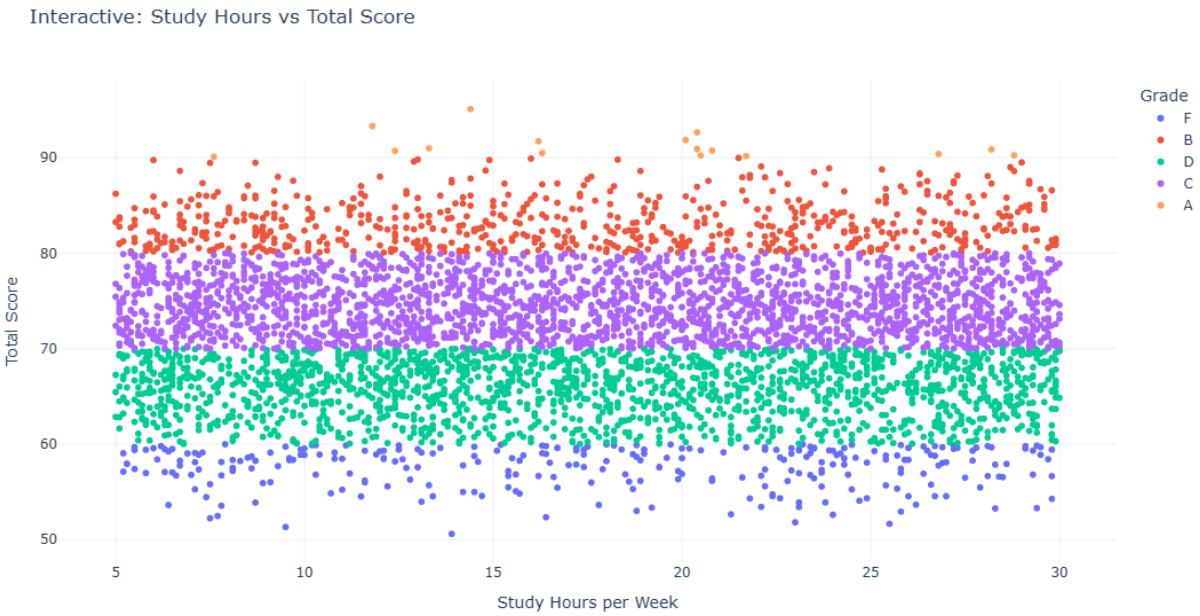


Figure 6.18a: Study hours vs

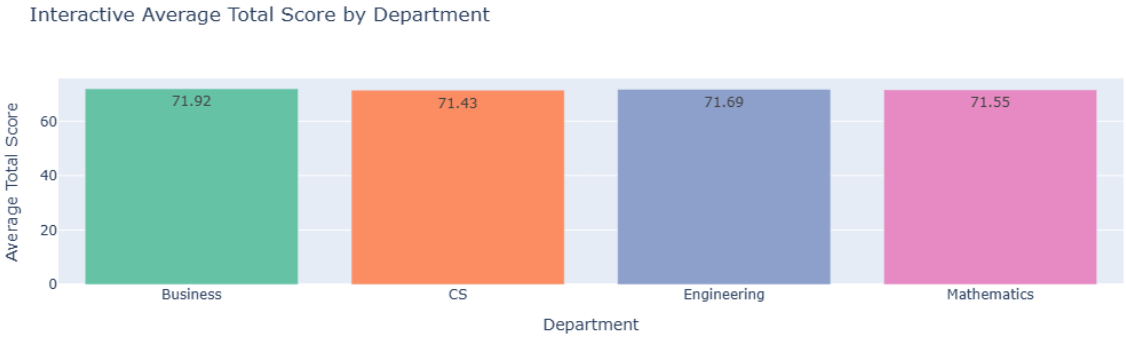


Figure 6.18b: Average Total Score by department (interactive Plotly bar chart, static export).

7. Key Insights

The guidelines require at least five insights; eight were derived.

1. **Assessment components matter most.** Projects_Score (0.60) and Final_Score (0.59) have the strongest correlations with Total_Score, followed by Midterm and Assignments.
2. **Performance is concentrated in the middle.** Most students earn C or D grades; only 16 students (0.32%) earned an A. The score distribution peaks around 70 to 75.
3. **Department has limited impact.** Average Total Scores range only from 71.43 to 71.92.
4. **Gender makes little difference.** Female and male averages are 71.64 and 71.66.
5. **Attendance is not a strong driver.** Its correlation with Total Score is about -0.01, and average attendance does not rise with grade.
6. **Study hours show almost no linear link with scores.** The correlation is also about -0.01 and the regression line is nearly flat.
7. **Stress and sleep differ only slightly across performance groups,** yet 313 students (6.26%) have both poor sleep and high stress, so subgroups can matter even when overall correlations are weak.
8. **Missing data was concentrated in one column.** Parent_Education_Level had 1,025 missing values (20.5%), retained as an Unknown category; family background (education and income) had little effect on scores.

8. Answers to the Key Analytical Questions

1. **Which academic factors relate most to overall performance?** Projects_Score (0.60) and Final_Score (0.59).
2. **How does performance vary across departments and gender?** Barely. Departments range from 71.43 to 71.92 and gender averages are 71.64 and 71.66.
3. **Do attendance and study hours show a meaningful relationship with performance?** No. Both have correlations of about -0.01 with Total Score.
4. **Are there groups needing closer analysis?** Yes: the 313 students (6.26%) with poor sleep and high stress, and the very small top-performer group (16 students).
5. **What other patterns help explain performance?** Academic assessments show clearer relationships with outcomes than behavioral or demographic factors, and subgroup analysis reveals patterns that overall correlations miss.

9. Recommendations and Next Steps

Recommendations

- **Focus on projects:** use more practical, project-based work, since projects have the strongest link with overall performance.
- **Support final exam preparation:** provide revision plans and practice assessments.
- **Identify struggling students early:** monitor students with consistently low grades.
- **Encourage consistent academic work:** give regular assignments, quizzes and feedback.

- **Monitor well-being:** track stress and sleep as supporting indicators, particularly for the poor-sleep, high-stress group.

Next steps

1. Build an early-warning system to identify students who may need support.
2. Track performance over time by comparing progress across semesters.
3. Explore other factors such as learning methods, motivation and family support.
4. Examine possible non-linear relationships between behavioral and academic factors.
5. Build a simple predictive model to identify factors associated with performance.
6. Collect more data from future semesters to validate and improve the findings.

10. Limitations

- **Correlation is not causation.** The findings describe associations, so recommendations such as increasing project work should be treated as hypotheses to test.
- **Built-in relationships.** Total_Score is made up of assessment components, so its correlation with those components is partly by construction. Grade is derived from Total_Score, so the Grade-to-score pattern is also expected.
- **Small groups.** The Excellent (Grade A) group has only 16 students, so its averages are unstable.
- **Descriptive methods only.** No hypothesis tests or predictive models were used, and linear correlation can miss non-linear patterns.
- **Self-reported behavior and dataset origin.** Study hours, stress and sleep are likely self-reported, and the near-zero effects plus small name pools (8 first names, 6 last names) suggest the data may be simulated, so results may not generalize to real student populations.

11. Conclusion

This project transformed 5,000 student records into meaningful insights through data cleaning, exploratory analysis, statistical analysis, visualization and interpretation. The dataset was clean after handling one missing-value column, and the analysis showed that assessment components, especially projects and final exams, are most closely associated with overall performance, while demographics, attendance, study hours, stress and sleep show weak overall relationships. Subgroup analysis added value by surfacing 313 students facing both poor sleep and high stress. The real value of data analytics is not just finding numbers, but discovering the story behind them.

Appendix: Compliance with Project Guidelines

Requirement	How it was met
Minimum 500 records	5,000 records
At least 10 features, mixed types	23 original columns (29 after cleaning), numerical and categorical
Data loading and overview	Shape, dtypes, head, tail, info, describe (Section 3)
Pre-processing	Missing values, duplicates, types, validation, outliers, derived columns, filtering, aggregation (Section 4)
Univariate, bivariate, multivariate EDA	Section 5
Groupby, pivot tables, correlation, statistical summaries	Sections 5.3 to 5.6
At least 10 visualizations	21 dedicated plus 2 in pre-processing (Section 6)
Titles, labels, legends, colours, subplots	Applied throughout; 2x2 subplot and pairplot included
At least 5 insights	8 insights (Section 7)
Markdown explanation throughout	Every step and chart interpreted in the notebook
Conclusion and recommendations	Sections 9 and 11
Original dataset source	Kaggle link in Section 1.5